$X = x$, corresponds to the combination of x heads in n tosses, that is $\binom{n}{x}$ ways of heads and remaining $n - x$ tails. Hence, the probability for each of those outcomes is equal to $p^x(1-p)^{n-x}$. Binomial theorem is suitable to apply when n is small number less than 30.

Definition 11.11: Binomial random variable

A discrete random variable X is called binomial random variable, if X is the number of successes in n-repeated trials such that

(i) the n- repeated trials are independent and n is finite
(ii) each trial results only two possible outcomes, labelled as 'success' or 'failure'
(iii) the probability of a success in each trial, denoted as p, remains constant.

Definition 11.12 : Binomial distribution

The binomial random variable X equals the number of successes with probability p for a success and $q = 1 - p$ for a failure in n-independent trials, has ***a binomial distribution*** denoted by $X \sim B(n, p)$. The probability mass function of X is

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, 2, ..., n.$$

The name of the distribution is obtained from the *binomial expansion*. For constants a and b, the binomial expansion is

$$(a+b)^n = \sum_{x=0}^{n} \binom{n}{x} a^x b^{n-x}$$

Let p denote the probability of success on a single trial. Then, by using the binomial expansion with $a = p$ and $b = 1 - p$, we see that the sum of the probabilities for a binomial random variable is 1. Since each trial in the experiment is classified into two outcomes, {success, failure}, the distribution is called a "bi"-nomial.

If X is a binomial random variable which follows binomial distribution with parameters p and n, the mean μ and variance σ^2 are

$$\mu = np \quad \text{and} \quad \sigma^2 = np(1-p)$$

The expected value is in general not a typical value that the random variable can take on. It is often helpful to interpret the expected value of a random variable as the long-run average value of the variable over many independent repetitions of an experiment. The shape of a **binomial distribution** is **symmetrical** when $p = 0.5$ or when n is large.

When $p = q = \frac{1}{2}$, the binomial distribution becomes

$$f(x) = \binom{n}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{n-x}, \qquad x = 0, 1, 2, ..., n.$$

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